Aditya Prabhakar

Roll No.:240107003
B.Tech - Chemical Engineering (Major)
Minor in Computer Science
Indian Institute Of Technology Guwahati

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.43 (Current)	2024-Present
B.Tech. Minor	Indian Institute of Technology, Guwahati	7 (Current)	2025-Present
Senior Secondary	CBSE Board	94.80%	2023
Secondary	CBSE Board	93.20%	2021

Experience

- **AYUSH Upchar** — *Aug. 2025 – Oct. 2025*
  Web Developer Intern — Remote
  - Conducted a comprehensive **website performance audit** using Google PageSpeed Insights, analyzing **Core Web Vitals** and identifying major performance gaps across **Desktop (score: 59 : Poor)** and **Mobile (score: 28 : Poor)**.
  - Quantified client-side inefficiencies including **1.5 MB+ unused JavaScript**, **45 KB unused CSS**, render-blocking resources causing **~390 ms (desktop)** and **~480 ms (mobile)** delays, and up to **~9.3 s of main-thread blocking time**.
  - Authored a detailed **performance optimization report** outlining actionable improvements such as **deferred script loading**, **critical CSS inlining**, **CDN-based image delivery**, **SVG optimization**, and **semantic HTML enhancements**, projecting **PageSpeed improvements to 75+ (Mobile)** and **90+ (Desktop) based on Page Speed Optimizer reports**.
  - Documented **SEO and accessibility enhancements**, including image alt-attribute optimization and structural markup improvements, targeting an **SEO score of 90+**, and presented recommendations to the web development team.

Projects

- **Velora Corporate Mobility Optimization Engine** — *Jan. 2026 - Mar. 2026*
  Kriti 2026, Inter Hostel Tech Competition, IIT Guwahati — Website | Backend Github | Engine Github
  - Engineered a **custom, high-performance C++ optimization engine** from scratch to solve the **Capacitated Vehicle Routing Problem with Time Windows (CVRPTW)**, utilizing a Solomon I1 constructive heuristic and an **Adaptive Large Neighborhood Search (ALNS)** metaheuristic to minimize operational costs **and maximize fleet utilization**.
  - Developed a **Real-Time Dynamic Insertion module** that leverages forward-time propagation to splice new ride requests into active routes in **under 10ms**, ensuring **100% employee coverage** while maintaining a worst-case global execution time of **111ms** for large-scale enterprise datasets **without disrupting existing schedules**.
  - Delivered **57.65% to 79.86% in operational cost savings** over baseline market pricing by deploying a full-stack solution featuring a **Django REST API**, an interactive **React.js dashboard** with road-snapped geometries, and a cross-platform **Capacitor-based mobile application for seamless fleet management and operational efficiency**.

- **Vogue Nation Portal** — *Oct. 2025*
  Alcheringa 2026, IIT Guwahati — Website | Github
  - Built a **full-stack event registration and team management platform** using **Django, SQLite, and Google OAuth 2.0**, supporting secure authentication for **35+ teams and 140+ participants (ongoing) during live registrations**.
  - Designed **relational data models** with the **Django ORM**, implementing **one-to-many team relationships** via **formsets and Django signals** to handle dynamic team creation, input validation, and relational integrity **reliably**.
  - Deployed the application using **Gunicorn and Whitenoise**, configuring static asset handling and production settings to deliver a **responsive, animated UI** built with **Tailwind CSS and GSAP** for live fest usage **during the event**.

- **UDGAM 2026 Website** — *Jan. 2026*
  UDGAM 2026, IIT Guwahati — Website | Github
  - Designed and developed the **official frontend website** for UDGAM 2026 using **React, React Router, Tailwind CSS, and JavaScript**, serving as the primary public-facing platform and handling navigation, layout, and interactions.
  - Implemented **responsive layouts and reusable UI components** using **flexbox, media queries, and component-based design**, ensuring consistent rendering and smooth user experience across desktop, tablet, and mobile devices.
  - Optimized frontend performance and code structure by maintaining a **modular React component architecture**, reducing unnecessary re-renders and improving component reusability, readability, and long-term maintainability.

Technical Skills

- **Programming**: Python, C++/C, JavaScript, TypeScript*
- **Databases**: SQL, SQLite, MongoDB*
- **Web Development**: HTML, CSS, React.js, Django, Node.js, Next.js*
- **Tools and Platforms**: Git, GitHub, Google OAuth, Gunicorn, Figma, Android Studio (Capacitor)

** Elementary proficiency*

Achievements

- **Institute Merit Scholarship**, Awarded **(₹2.08 Lakhs)** for **highest academic standing** in the department. — *2025*